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Ginger (Zingiber officinale)

ORIGIN AND DISTRIBUTION

Ginger originated in tropical Asia but it is now grown all over the tropics. West Africa is an important exporter of the crop, Sierra Leone being the leading producer.

VARIETIES OF GINGER

At present only two varieties are grown in Nigeria. These are: -

- 1) Black ginger called Monkey-finger or 'Katsina Birni'.
 - 2) Yellow ginger called Elephant-foot or 'Taffin Gima'.
- These varieties are very good in ginger oil yield. The most popular and recommended variety in Sierra Leone is the yellow type.

AREA OF PRODUCTION

Ginger is a shallow rooting tropical crop grown in many parts of tropical Asia, Indonesia and Sierra Leone. In Nigeria, the cultivation of ginger started in 1927 and by 1977 Nigeria was one of the first five largest exporters of ginger in the world. The main area of production is around Jekesa District in Ibadan state.

ECOLOGICAL REQUIREMENT

Ginger requires moderate to heavy rainfall above 1000 mm per annum. The rainfall should be well distributed throughout the growing period. It grows best in lowland forest areas under shade.

Temperature from 23°C - 35°C is adequate. The soil should be a deep loam, rich in humus and well-drained. Sandy clay loam is ideal. Best yields are obtained from soils that are rich in organic matter.

LAND PREPARATION

To produce a well-shaped rhizome a good soil tilt is required. Ginger can be grown on flat ridges or beds. Beds are however preferred under rainfed conditions. If production is by the use of irrigation, it should be planted on ridges. Beds are made 1 metre wide.

PLANTING MATERIALS

Ginger is propagated by pieces of rhizome 2.5-5 cm long containing one or two eyes.

Planting

In Nigeria, ginger is planted from March to April in the rainforest zone of the south and by April to May in the Savannah zone of Eastern Nigeria in the north. The propagated pieces of rhizomes are planted in shallow holes 5-7.5 cm deep on the beds at a distance of 20 cm along the beds and 20 cm across.

CROP MAINTENANCE

(i) Mulching:

Heavily grass mulching should be done after planting. This is in addition to adding organic manure to the soil on decomposition.

encourage early and even sprouting
(c) it keeping weed growth.

(c) It checks erosion including raindrop and erosion
(d) It prevents the seeds from drying up.

2. FERTILIZER APPLICATION

Ginger responds very well to the application of manure, 25-30 tonnes of compost or farmyard manure per hectare is recommended. This should be applied at planting just before mulching. Inorganic manure that is fertilizer can be used also. After about 6-8 weeks of planting, apply NPK 15:15:15 at the rate of 350-400 kg/ha. It is recommended that the fertilizer be applied in two doses, the first at 6-8 week after planting and the second 12-14 week after planting, in heavy rainfall area. This is to reduce losses through leaching.

3. WEEDING

The farm should be weeded twice or three times as the need arises. Ginger cannot stand weed competition. Pre-emergent herbicides may be used. Oxidation at 1.5 kg a.i./ha or chlorothal dimethyl at 12 kg a.i./ha may be used. They could be applied 1-3 days after planting to give adequate weed control.

HARVESTING OF GINGER

Ginger is usually ready for harvest in 9-10 months. It requires at least 32 weeks of field life before harvesting. When the leaves turn yellow or when

1. Pre-harvesting Digging up the rhizomes are ready for harvest. They should then be dug up with hoes, mattocks, spades or even with the digging iron rod or stick and collected. The roots are then removed.

YIELD

Yields of up to 20-30 t/ha are possible under improved cultural management. Care should be taken not to damage the rhizomes or injure them as that will reduce the quality. An average yield of 750 kg/ha is obtained in Sierra Leone. The potential of the best variety is 2, 241 kg cured ginger per ha per annum.

STORAGE

Ginger is stored in pit. Such a pit should be covered with a roof. The rhizomes are loosely packed in the pit up to a height of 10-15 cm and the pit covered with wooden planks leaving a small gap of about 10 cm between the planks and the rhizomes for air circulation. The plant is then covered with mud. Ginger can also be stored in a cool room under sand beds. The crop may be left unharvested in the field to avoid storage problems provided there is no disease or pest infestation.

PROCESSING OF GINGER

1. Peeled or scraped ginger: This is ginger with the outer layer removed. The rhizomes are soaked in water overnight before removing the outer skin. The soaking is done to facilitate peeling.

ing the rhizomes. Some should be taken not to is where the essential oil of ginger is located. The peeled ginger is then dried.

2 Coated or unpeeled ginger: - This is where the ginger is washed and sun dried to reduce the moisture content without removing the cap or skin.

3 Black ginger: This is ginger which is dried after soaking for 10-15 minutes in boiling water with or without peeling before drying. This soaking is to kill the ginger. Peeled rhizome can be soaked in 2% clear lime solution for about 6 hours before drying. This called Blacked ginger.

Dried ginger is washed and ground into powder to form ground ginger.

ECONOMIC IMPORTANCE OF GINGER

- Ginger is used commercially as a spice in the confectionery and mineral water manufacturing industries.

- It has a penetrating flavour.

- It is used for culinary purposes and for the preparation of preserves, pickles, puddings, beverages, Candy and other food forms.

- In the confectionery industry, it is used for making Pies, biscuits, cakes and ginger bread.

- Ground ginger is a constituent of curry powder and other spice, seasoning and flavouring mixture.

- Ginger is used for medical preparations.

- The essential oil and oleoresin are used in perfumery.

PETTS AND DISEASES

• So far the incidence of petts has been minimal. Because of the low incidence of petts, not much research effort has been put into that area.

Diseases of Ginger

1. Black rot: The leaves of diseased plants turn yellow and wither and the rhizome disintegrates. The disease is common under wet conditions. Control :- Control is achieved by removing infected plants and treating the soil with lime or sulphate of iron.

2. Vermicularia Zingiberæ: This disease affects the leaves and prevents rhizome development. Control :- Spraying with Bordeaux mixture gives effective control.

YAM (Dioscorea spp)
ORIGIN AND DISTRIBUTION

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The white yam (*Dioscorea rotundata*) is grown on a greater hectarage and produced in larger quantities than any other kind of yam in the world. It originated in West Africa and remains the predominant kind of yam grown there. It is characterised by firm white tuber flesh, heart-shaped leaves with pointed tips, and spiny stem which is circular in cross section.

The water yam (*Dioscorea alata*): It originated in South-east Asia, and first reached Africa at the East African coast. By the nineteenth century it had been introduced to West Africa. The tuber flesh is generally white but loose in texture. The stem has four or more rows of warts running its entire length. The leaf petioles also have warts. The leaves are heart-shaped and pointed, but are usually larger than those of the white yam.

The yellow yam (*Dioscorea cayenensis*) originated in West Africa and is not grown very extensively outside the area. The tuber flesh is usually yellow while the corm attached to the tuber is larger than that of most other yams. The leaves are dark green in colour and usually broad. The yellow yam requires a longer season for maturity than most other yams.

The trifoliate yam (*Dioscorea dumetorum*) originated in tropical Africa, and is today cultivated extensively only in West Africa. It differs from most of the other kinds of yam in having trifoliate leaves and hairs on the stems and leaves. Unlike the previous three kinds of yam, which have

to the right, the trifoliate yam twisted to the left, i.e. it approached the support keeping to the left. The tuber is large, coarse, irregularly shaped and somewhat bitter, so that it requires long periods of steeping in water and boiling before it is eaten.

The Chinese yam (Dioscorea esculenta) originated in South-East Asia. It was first cultivated in China several centuries ago and today it is extensively grown mainly in South-East Asia, the Pacific and the West Indies. It is relatively new to Africa, and its cultivation there remains quite limited. The stem twists to the left and unlike most other kinds of yam, each plant produces a cluster of several small tubers rather than one or two large ones. The tuber has a higher sugar content than those of other yams and is therefore sweet-tasting.

The aereal yam (Dioscorea bulbifera) originated in both Africa and Asia. The plant characteristically produces numerous aereal tubers or bulbils which are used for food. It also produces some edible underground tubers.

VARIETIES OF YAMS

Yams are classified on species level. They are obtained from several species of the genus Dioscorea, a very large genus of climbing tropical plants. They are monocotyledonous and the family Dioscoreaceae is derived from the family Liliaceae.

The classes include:

- a. White yam (Dioscorea rotundata)
- b. Yellow yam (Dioscorea cayenensis)

on the gam species. D. rotundata, D. caryensis and D. alata creep to the right when approaching a support and therefore move in a clockwise direction going upward. D. thurmetorum, D. bulbifera and D. esculenta creep to the left when approaching a support.

The stems in most gams are more or less circular in cross-section but in D. alata the stem is usually star-shaped in cross-section. Spines are commonly found on the stems of the whole gam, yellow gam, trifoliate gam and Chinese gam, while hairs are present on the stem of the trifoliate gam.

For most kinds of gams, the leaves are simple, net-veined, devoid of hair and more or less heart shaped. The leaves are generally green although leaves of some cultivars of D. alata may be purple, especially when young. Leaf arrangement on the stem is either opposite or alternate. Dioscorea thurmetorum differs from most other gams in having trifoliate leaves with hairs on both the petiole and the lamina.

Gams are usually dioecious, producing male and female flowers on different plants; although a few cases of male and female flowers being borne separately on the same plant have been reported. The gam fruit is a capsule with three lobes and a maximum of two seeds per lobe. The seed usually goes through a dormancy period of 3-4 months before it can germinate.

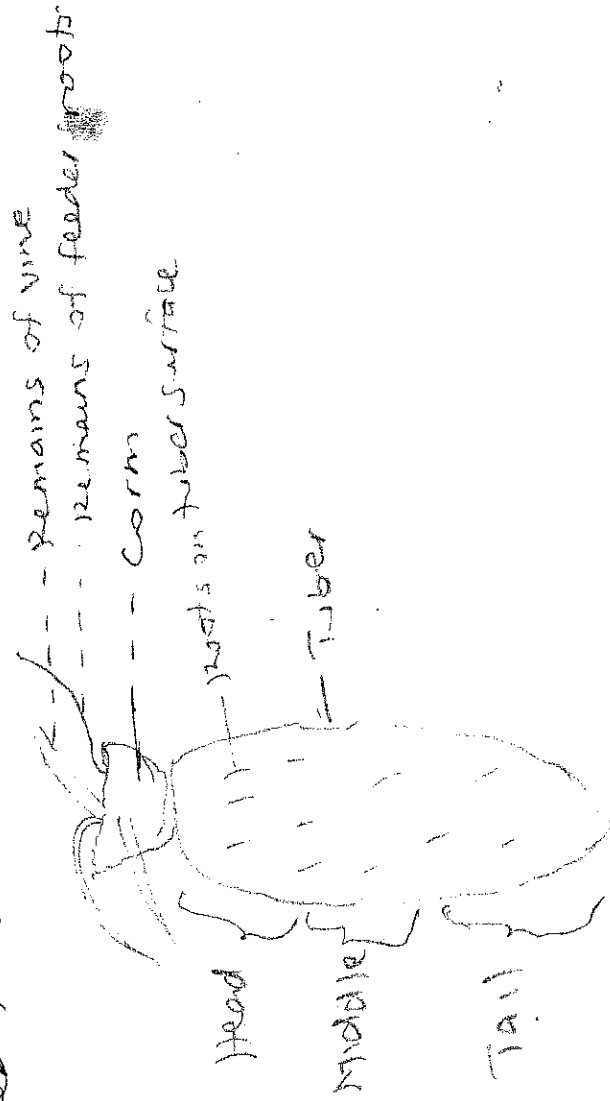
Flowering and seed production in gams are extremely irregular. Some species such as D. alata flower only rarely, while others such as D. caryensis produce only male flowers and the whole gam.

- c. Water Yam (Diocorea alata)
- d. Trifoliate yam (Diocorea dumetorum)
- e. Chinese yam (Diocorea esculenta)
- f. Aerial yam (Diocorea bulbifera)

BOTANY OF YAMS

Yams have a fibrous root system conformed mostly to the top 30 cm of soil. In species where the stem is very spiny, some spines may also be present on the roots. In many species, a few thin roots can also be found on the body of the tuber.

The yam tuber is neither a stem nor a root tuber. The tuber itself is attached to a corm which in turn is attached to the main roots and stem of the plant.



Indeed, the corm serves as the central point of origin of the roots, stem and tuber of the plant. The corm is woody and possesses dormant buds, but the tuber itself is devoid of buds. The part of the tuber nearest the corm is called the head, while the most distant part is the tail. A few thin roots may occur on the tuber surface.

The yam stem is a thin twining structure. The

produce seeds. Even for species where flowering is common, a sizeable proportion of the plant on the field compete their annual growth cycle and die without producing any flower where flowers are produced, pollination is poor and ovule abortion is high.

ECOLOGICAL REQUIREMENTS

Jams grow best at temperatures of 25-30 and their cultivation is essentially restricted to the tropics. Their growth is considerably retarded at temperatures below 20°C. Since they require 6-9 months to mature on the field, it is even that the rainy season lasts at least as long, in supplementary irrigation can be provided. A well-distributed rainfall (for water supply) of 1500 mm per annum is adequate for jam production. However, dry spells of up to a month during growing season need not severely reduce yields. Tuber formation in jams is promoted by short conditions but low light intensity or shade during the season tends to reduce jam yields severely. Compared with other tropical tuber crops, it requires soil of high fertility. Loose loamy or are best as heavy clays tend to be waterlogged and result in tuber rot and difficult harvest.

LAND PREPARATION

Land preparation for planting jams and the same as that already described for planting cassava. After the land has been cleared, the soil is made ridges or mounds. Occasionally jams are planted

on the flat after ploughing, or in holes dug in unploughed land. However, yams grown on the flat or in holes yield lower and are more difficult to harvest than those grown on ridges and mounds.

PLANTING MATERIALS

Yams are normally propagated using tuber material. The piece of tuber that is used for planting is called a sett. It may be a small whole tuber, or a piece of tuber derived from the head, middle or tail region of a larger tuber. As a rule, the head portion of the yam tubers sprout more readily than the middle and tail portions. On the other hand, whole tubers have no cut surface and are therefore less prone to rotting when planted than head, middle or tail portions.

As a rule, the larger the sett weight used, the greater the tuber yield obtainable from the resulting plant. This is because large setts sprout more readily, establish more quickly, and give rise to more vigorous plants with a larger total leaf area. Very small setts are therefore disadvantageous because they result in very small tubers which have low market value. However, very large setts are also disadvantageous because they result in low multiplication ratio. Very large setts are therefore wasteful of planting materials.

CROP ESTABLISHMENT

Sowing is normally with the early rains but in areas where there are heavy soils or loose alluvial soils, yams can be sown at the end of the rains around November - December otherwise sowing should be from February - March.

Planting can be with yam sets or seed yams. The sets should be treated against yam beetle by rolling them in DDT in 2.5% dust before planting.

The spacing is 1m x 1m, 1.3m x 1.3m, 2m x 2m as the case may be. The greater spacing is used where there is mixed cropping.

CROP MAINTENANCE

Weed control

Most of the farmers weed their fields 3 or 4 times during the season, using ~~hops~~ or machetes. Experiments have shown that yams are most sensitive to weed competition between emergence and the time of tuber initiation in the third or fourth month after emergence. It is therefore essential that adequate weed control is maintained at least during this period.

The use of herbicides to control weeds in yam is not very widespread in Africa, but it is increasing. The most commonly recommended herbicides are diuron (3-3.5 kg/ha), atrazine (3.5 kg/ha) and ametryne (3-4 kg/ha). These are all effective on weeds before they emerge and may therefore be applied soon after the yams are planted.

2. Staking

This is the provision of same support for the yam vines. Such staking materials as the stems of grasses, bamboos, raffia and palm fronds and the stems of erect plants are used. Crops are also used in some cases. These are tied to trees and the vines directed to trail up the ropes. Yield can be reduced by about 50% if there is no staking.

3. Fertilizer Application

Compost and farmyard manure, produced and used at the village level, have been found to result in improved yam yields. Commercial fertilizers are also in common use. Apparently yams respond well to nitrogen and potassium fertilizers, but very little to phosphorus. The exact fertilizer formulation and quantity used will depend on the local soil conditions. When inorganic fertilizers are used, it is recommended that the dose be split; the first application is made about a month after emergence, while the second is made during tuber bulking about 3 months after emergence. The fertilizer commonly applied is NPK 15-15-15 at 400 kg/ha.

HARVESTING

Two kinds of practice exist with respect to the timing of the yam harvest.

In double harvesting, each yam plant is harvested twice. A first harvest is done at about 5 months after emergence. During this harvest, the farmer ~~carefully~~ carefully removes the tuber without damaging the plant's roots. The tuber is removed and the roots are covered up again with earth. The plant continues to grow and later produces a new tuber. After the plant has finally died at the

end of the season, the second harvest is done
tuber recovered at the second harvest is from
in texture and more irregular in shape than
that from the first harvest.

Single harvesting involves waiting until the
end of the season when the vines have died, be-
fore harvesting the tubers. Each plant is therefore
harvested just once.

The mechanical harvesting of tubers is not a
common practice at present, but it is certainly
desirable for the future.

YIELDS

The average yields of fresh yam tubers in
Africa are about 10 t/ha. Higher or lower yields
may be obtained depending on the cultivation
and on the level of management. Bigger sets
produce higher yields than smaller ones. In
varieties produce one large tuber or at
most, two or three tubers per plant. High
yields of up to 15-25 t/ha are possible on
good soil which has been fertilized, if there
adequate rainfall.

STORAGE

Most of the yams produced in Africa are stored
in the form of the fresh tuber. There are three
common methods of storing the tubers:

① In barns ② On platforms and ③ Underground.

The barn is the commonest form of yam storage
in West Africa. It is erected in an open place and

Consists ~~of~~ essentially of a series of vertically-oriented poles to which the yams are tied with rope.

In platform storage, the yams are laid horizontally on an elevated platform.

In underground storage the tubers are placed together in a large ditch and then covered with soil or dry vegetation.

Whichever method of yam storage is adopted, it is essential that the tubers should be well aerated and well shaded. In addition, the stored tubers should be inspected frequently so that rotting ones can be removed, and the sprouts can be removed from those that begin to sprout.

ECONOMIC IMPORTANCE OF YAMS

The freshly harvested yam tuber consists of about 70% water, 25% starch, 1-2% protein and only traces of sugars and vitamins.

The most common forms of yam consumption are as boiled yam and pounded yam. Boiled yam is produced by boiling the tubers (before or after peeling) in water.

It is consumed with oil, stew or vegetables.

Pounded yam is made from boiled yam by pounding the boiled yam in a mortar to produce a thick paste.

Processed forms in which yam can be stored are also found in Africa. The most common is yam flour. This made by drying thin slices of the peeled tuber and milling the dried pieces to produce a flour which can be stored dry for several months. When needed for consumption,

the flour is stirred in boiling water over a fire to produce a paste. The paste is similar to powdered gum, and is eaten in the same way. Other preferred forms of gum include gum chips, which are used as a snack, and gum flakes. The flakes when stirred with boiling water, reconstitute into a product identical to powdered gum.

PESTS AND DISEASES OF GAMS

PESTS OF GAM

- (a) Gam beetles (Heteroligus spp) are a major pest of gum, particularly in West Africa. The beetles do considerable damage to the tubers in the field, digging deep holes on the body of the tuber.
- Control: - They are best controlled by dusting or planting seed with insecticides such as aldrin or lindane (Graminivore).
- Gam shot beetle: The larvae of this beetle attack young gum shoots, causing stunted growth.
- Control: - During dry weather it can be pushed off by hand.
- (b) Gam edworm or nematode: - This pest attacks tubers in the soil when the gums are sown, causing them to rot. Attached tubers become brown, soft, and are unfit for consumption.
- Control: Crop rotation is the main control.
- (c) Crickets (Gryllus spp): These are common pests of gum. They can cut the vines or they go down and damage the tubers, though their damage is not as destructive as that of gam beetles.

Control: ① By using Aldrin 2.5%

② Hand picking of the crickets can also be an effective control.

③ Yam leaf beetles: - These attack the vegetative part of the crop often causing stunted growth.
Control: It can be by spraying with Dieldrin or dusting with either Aldrin 2.5% or Vetox 5.

DISEASES OF YAM

The main diseases affecting yams are: -

① Tuber rots: - caused mostly by fungi, are among the most serious diseases of yams. The rots may infect the tubers either in the field or during storage.

Control: ① It can be controlled by avoiding tuber wounding during harvest and by good soil aeration in the field.

② Anthracnose disease is particularly severe when it attacks Dioscorea alata. Leaf necrosis and death of the entire plant occur, so that the yield losses are considerable.

Control: The use of resistant varieties such TDA 291 and Apommapondem, appears to be the best control measure.

③ Witch broom: - This is a fungal disease which causes dwarfed vines that produce a large number of branched of almost the same height. The leaves are small and chlorotic and the infected stems bear long band streaks.

Control: - The best control is to destroy and burn all